PANJAB UNIVERSITY CHANDIGARH

(Estted. under the Panjab University Act VII of 1947-enacted by the Govt. of India)

FACULTY OF ARTS

SYLLABI

FOR

DISASTER MANAGEMENT
(ADD-ON COURSE)
CERTIFICATE, DIPLOMA
AND
ADVANCED DIPLOMA
EXAMINATIONS, 2015

PANJAB UNIVERSITY, CHANDIGARH

Outlines of tests, syllabi and courses of reading for Disaster Management (Add-on Course) for the examinations of 2015.

CERTIFICATE COURSE

Course Structure

Course Code	Title of the Paper	Max. Marks
CCDM I	Fundamentals of Disaster Management Theory : 75 Practical : 25	100
CCDM II	Disasters in India Theory : 75 Practical : 25	100

CCDM I: Fundamentals of Disaster Management

Maximum Marks : 100
Theory Paper : 75
Practical : 25
Time : 3 hrs.

Objectives:
To introduce the students about basic concepts of various types of disasters and educate them on basic disaster management strategies.

Unit-I

- Definition and Concept of Hazard, Risk, Vulnerability and Disaster
- Types and classification of Disasters

Unit-II

- Nature Induced Disasters:
  Earthquakes, Floods and Cyclones.
  Avalanches, Forest Fire and Tsunami.

Unit-III

- Human Induced Disasters:
  Nuclear, Chemical and Industrial Disasters
  Global warming; Biological Disasters; Epidemics

Unit-IV

- Disaster Management: Meaning, Concepts, Approaches, Principles, Objectives and Scope

- Essentials of Disaster Management; Institutional and Individual's responsibilities during risk reduction, preparedness, response and recovery phases

Pedagogy:
Focus should be on observation, assessment and careful analysis of hazard prone areas to comprehend their vulnerability to disasters.

Note:

a) There shall be one compulsory question containing 15 short type questions covering the whole syllabus. The student shall attempt any ten parts in about 25-30 words each. Each part shall carry 1.5 marks (15 marks).

b) The whole syllabus shall be divided into 4 units. Eight questions will be set from the whole syllabus, two from each unit. The student will be required to attempt one question from each unit. Each question will carry 15 marks. This will be in addition to the compulsory question.

Practical:

1. A visit to a school for vulnerability assessment.
2. A visit to any Government Hospital to know their disaster management plan
3. A visit to one slum area in the City to assess and analyze risk.

Note:

1. Based on these visits, students should prepare a report containing the vulnerability assessment of a village, disaster management plan of a hospital and risks of disasters in a slum.

2. The report should be submitted two weeks before the exam. The report will be evaluated by a Board of Examiners consisting of the teacher teaching the course and one faculty member from the allied discipline.

List of Readings:

Essential Readings:

1. B.K. Khanna: *Disasters: All You Wanted to Know About*, New India Publishing Agency, New Delhi, 2005.

2. Bryant Edwards: *Natural Hazards*, Cambridge University Press, U.K, 2005.

3. Chakraborty, S.C.: *Natural Hazards and Disaster Management*, Pragatishil Prokashak, Kolkata, 2007.

Further Readings:

1. Sahni, Pardeep et.al. (eds.) : *Disaster Mitigation Experiences and Reflections*, Prentice Hall of India, New Delhi. 2002.

2. Kapur, A.*: Disasters in India: Studies of Grim Reality,* Rawat Publications, Jaipur, 2005.

CCDM II: Disasters in India

Maximum Marks	: 100
Theory Paper	: 75
Practical	: 25
Time	: 3 hrs.

Objectives:
To introduce the students about the various disasters that have occurred in the recent past and their management in India.

Unit-I

- Regional Profile of India based on Earthquakes, Droughts, Floods and Cyclones.

Unit-II

- Disaster Management Framework in India
- National Level, including National Disaster Management Authority.
- State Authorities, Local Groups and Committees

Unit-III

- Case Studies: Bhopal Gas Tragedy, 1984; Orissa Super Cyclone, 1999; Kashmir Earthquake, 2005, and Kosi River Flood, 2009

Unit-IV

- Role of NGOs, Charitable Trusts, Army and Police, and Educational Institutions in Disaster Management

Pedagogy:

Visuals and DVDs on various disasters in India shall be shown to students.

Note:

a) There shall be one compulsory question containing 15 short type questions covering the whole syllabus. The student shall attempt any ten parts in about 25-30 words each. Each part shall carry 1.5 marks (15 marks).

b) The whole syllabus shall be divided into 4 units. Eight questions will be set from the whole syllabus, two from each unit. The student will be required to attempt one question from each unit. Each question will carry 15 marks. This will be in addition to the compulsory question.

Practical Exercises: A field visit of the students will be organized by the College to some nearby site of any recent disaster/ event or ecological destruction or reconstruction and each student would prepare a Field Report based on his/her experiences.

Note: The Field Work based report of 25-30 pages prepared by the students will be submitted two weeks before the theory examination. This will be evaluated by a Board of Examiners consisting of the teacher teaching the course and one faculty member from the allied discipline.

List of Readings:

Essential Readings

1. Kapur, A.: *Disasters in India: Studies of Grim Reality*, Rawat Publications, Jaipur; 2005.

2. Paraswamam, S. and Umikrishnan, P.V., *India Disaster Report*, Oxford University Press, New Delhi, 2000.

3. Reddy, K.R and Others (eds).: *Natural Hazards and Disasters*, Department of Geography, Sri Krishnadeveraya University, Anantapur, A.P., India, 2009.

Further Readings:

1. Sharma, V.K. (ed.): *Disaster Management*, Indian Institute of Public Administration, New Delhi, 1994.

2. Bryant Edwards: *Natural Hazards,* Cambridge University Press, U.K, 2005.

DIPLOMA COURSE

Course Structure:

Course Code	Title of the Paper	Max. Marks
DCDM I	Remote Sensing in Disaster Management Theory : 75 Practical : 25	100
DCDM II	GIS in Disaster Management Theory : 75 Practical : 25	100

DCDM-I: Remote Sensing in Disaster Management

Maximum Marks : 100
Theory Paper : 75
Practical : 25
. Time : 3 hrs.

Objectives:
To introduce the students to the basic concepts and application of Remote Sensing in Disaster Management

Unit - I

- **Basics of Remote Sensing:**

  Introduction, Definition, Nature, Development, and Limitations.
  Role of Remote Sensing in Disaster Management

Unit - II

- **Basics of GPS**:

  Concept, Definition, Elements and Factors of GPS.
  The Mechanism of GPS for Disaster Management.

Unit - III

- **Remote Sensing Applications in Natural Disasters:**

  Earthquakes
  Land Slides
  Floods

Unit - IV

- **Remote Sensing Applications in Human Induced Disasters**:

  Fire
  Nuclear
  Chemical

Pedagogy:

Basic Fundamentals of Remote Sensing and GPS to be introduced by Aerial photograph interpretation. Training in all aspects of GIS & Remote Sensing and its real world applications.

Note:

a) There shall be one compulsory question containing 15 short type questions covering the whole syllabus. The student shall attempt any ten parts in about 25-30 words each. Each part shall carry 1.5 marks (15 marks).

b) The whole syllabus shall be divided into 4 units. Eight questions will be set from the whole syllabus, two from each unit. The student will be required to attempt FOUR questions selecting one from each unit. Each question will carry 15 marks. This will be in addition to the compulsory question.

Practical :

Field Trips (one week practical attachment programme visit to Emergency sites, man-made and natural hazard locations)

To visit the sites of previous or current disaster as well as agencies dealing with disasters or disaster prone areas/districts in various ways.

Note:

a) Based on these visits, students should prepare a report.

b) The report should be submitted two weeks before the exam. The report will be evaluated by a Board of Examiners consisting of the teacher teaching the course and one faculty member from the allied discipline.

List of Readings:

Essential Readings:

1. R.B. Singh: *Natural Hazards and Disaster Management,* Rawat Pulications Jaipur, 2009.
2. Saumitra Mukherjee: *Remote Sensing Applications in Applied Geosciences* Manak Publishers Pvt. Ltd. 1999.
3. A.M. Chandra and S.K. Ghosh: *Remote Sensing and Geographical Information System*, Narosa Pulishing House, New Delhi, 2006.
4. Victor Mesev: *Integration of GIS and Remote Sensing*, John Wiley and Sons Ltd. 2007.
5. Thomas M. Lillesand, Ralph W. Kiefer, Jonathan W. Chipman : *Remote Sensing and Image Interpretation*, Wiley India Pvt. Ltd. New Delhi.-2003.
6. Robert A. Schowengerdt: *Remote Sensing-Models and Methods for image Processing*, Academic Press- California U.S.A.-2006.

Further Readings:

1. Shailesh N. and S. Zlatanova (eds.): *Remote Sensing and GIS Technologies for Monitoring and Prediction of Disasters,* Springer-Verlag, Berlin, Heidelberg, 2008.

2. Nirupama, Slobodan P. Simonovic: *Role of Remote Sensing in Disaster Management*, University of Western Ontario. Dept. of Civil and Environmental Engineering.-2002.
3. Warner, T.A., M. D. Nellis and G. M. Foody: *The SAGE Handbook of Remote Sensing,* SAGE Publication Ltd., London, 2009.
4. Jonathan Li, S. Zlatanova and A. Fabbri (eds.): *Geomatics Solutions for Disaster Management*, Springer, Verlag, Berlin, Heidelberg, 2007.

DCDM-II: GIS in Disaster Management

Maximum Marks: 100
Theory Paper : 75
Practical : 25
Time : 3 hrs.

Objectives: To introduce the students about the basic concepts and application of GIS in Disaster Management.

Unit - I

- **Basics of GIS**:

  Introduction, Definition, Nature, Development.
  Components of GIS
  Application of GIS in Disaster Management

Unit - II

- **Geo-informatics Perspectives in Disaster Management:**

  Structure for Storing Disaster Data.
  Spatial Analysis of Disaster.

Unit - III

- **Disaster Mitigation Mapping:**

  Natural Hazard Analysis and Mapping
  Man Made Hazards and Mapping

Unit - IV

- **Risk and Vulnerability Assessment:**

  Warning and Evacuation
  Consolidated Preparation for next Disaster

Pedagogy:

Basic Fundamentals of GIS to introduce Disaster Management.

Note:

a) There shall be one compulsory question containing 15 short type questions covering the whole syllabus. The student shall attempt any ten parts in about 25-30 words each. Each part shall carry 1.5 marks (15 marks).

b) The whole syllabus shall be divided into 4 units. Eight questions will be set from the whole syllabus, two from each unit. The student will be required to attempt FOUR questions selecting one from each unit. Each question will carry 15 marks. This will be in addition to the compulsory question.

Practical

Practical Disaster Exercises and online activities.

Note:

a) Based on these visits, students should prepare a report.

b) The report should be submitted two weeks before the exam. The report will be evaluated by a Board of Examiners consisting of the teacher teaching the course and one faculty member from the allied discipline.

List of Readings

Essential Readings:-

1. Dr. M.A.Siddiqui : *Introduction to Geographical Information System*, Sharda Pustak Bhawan Allahabad.-2005.
2. Victor Mesev : *Integration of GIS and Remote Sensing*, John Wiley and Sons Ltd.-2007.
3. Dr. K. Elangovan : *GIS- Fundamentals, Applications and Implementations*, New India Publishing Agency- Pitampura- New Delhi.-2006.
4. Lan Heywood, Sarah Cornelius, Steve Carver, Srinivasa Raju: *Introduction to Geographical Information System,* Dorling Kindersley India Pvt. Ltd.2006.
5. Kang Tsung Chang : *Introduction to Geographic Information Systems,* Tata Mc Graw-Hill, Publishing Company Ltd.- New Delhi.2006.
6. Greg Bankoff Frerks, Dorothea Hilhorst: *Mapping Vulnerability: Disaster Development and People*, Earthscan Publications Ltd., 2004.
7. Monmonier Mark: *Cartographies of Danger: Mapping Hazards in America*, University of Chicago Press Publication.1997.

Further Readings:

1. Van Oosterom P, S. Zlatanova and E.M. Fendel: *Geo-information for Disaster Management,* Springer, Heidelberg, 2005.
2. Hadeki Kaji : *GIS for disaster management: proceedings*, United nations centre for regional development, 1997.
3. Sudha Maheshwari :*Disaster damage estimation models: Data needs vs. ground reality,* Rutgers The State University of New Jersey New Brunswick, 2007.
4. Sanjay Rana, Jayant Sharma : *Frontiers of geographic information technology*, Birkhäuser, Science 2006.
5. C.P.Lo and Albert K.W.Yeung : *Concepts and Techniques of Geographic Information Systems* Prentice Hall of India, New Delhi, 2005.

ADVANCED DIPLOMA

Course Structure:

Course Code	Title of the Paper	Max. Marks
ADCDM V	Insurance and Legal Aspects in Disaster Management Theory : 100	100
ADCDM VI	Project Report Based on Field Work. Practical : 100	100

ADCDM-V: Insurance and Legal Aspects in Disaster Management

Maximum Marks : 100
Theory Paper : 100
Time Allowed : 3 Hours

Objectives:

To introduce the students about the basic concepts and application of Insurance and Legal Aspects in Disaster Management

Unit – I

- Role of Human Rights in Disaster Management
- The Disaster Management Act-2005
- Basic Introduction of The Essential Services Maintenance Act-1981
- Role of PIL and Public Activism

Unit – II

- Role of the Union Government and Planning Commission in Disaster Management
- Role of State Governments in Disaster Management
- Role of Local Governments and NGOs in Disaster Management

Unit – III

- Administrative Reform Commission-2 and Disaster Management.
- Insurance and Disaster Management.
- Insurance, Assurance and Re-insurance.

Unit – IV

- Role and importance of Finance in Disaster Management.
- Role of World Bank in Disaster Finance and Risk Reduction.
- Financing Disaster Risk in India

Pedagogy:

Basic Fundamentals of Insurance and Legal Aspects in Disaster Management to be introduced to the students.

Note:

a) There shall be one compulsory question containing 15 short type questions covering the whole syllabus. The student shall attempt any ten parts in about 25-30 words each. Each part shall carry 2marks (20 marks).

b) The whole syllabus shall be divided into 4 units. Eight questions will be set from the whole syllabus, two from each unit. The student will be required to attempt one question from each unit. Each question will carry 20 marks. This will be in addition to the compulsory question.

Essential Readings:-

1. Andrew Waite: *Environmental Law: Hand Book*, London, Butterworths, 1984.
2. David Fisher: *Law and Legal Issues in International Disaster Response: A Desk Study,* International Federation of Red Cross and Red Crescent Societies, 2007.
3. Michael H. Hoffman: *International disaster response laws, principles and practice: reflections, prospects and challenges,* 2003.
4. S.Shanthakumar: *Introduction to Environmental Law*, Wadhwa & Co, Nagpur, 2007.
5. Ramesh Kumar Arora, Rajni Goyal : *Indian Public Administration: Institutions and Issues* - New Age International, 1995.
6. Arun Ray: *National Human Rights Commission of India: Formation, Functioning, and Future Prospects, Atlantic* Publishers & Dist, 2004.
7. Showick Thorpe & Edgar Thorpe : *The Pearson General Studies Manual 2009*, Pearson Education India, 2009.
8. Andrew Hiles : *The Definitive Handbook of Business Continuity Management,* John Wiley & Sons, 2010.
9. Dharmendra S. Sengar: *Environmental Law*, PHI Learning Pvt. Ltd., 2007.
10. V.V.Sople : *Logistic Management* , Pearson Education India, 2012.
11. P.B. Sahasranaman : *Handbook of Environmental Law* , Oxford University Press, 2009.

Further Readings:

1. Howard Kunreuther & Richard J. Roth: *Paying the Price: The Status And Role Of Insurance Against Natural Disaster,* Joseph Henry Press, 1998.
2. Howard Kunreuther: *Disaster Insurance Protection: Public Policy Lessons*- Wiely Press, 1978.
3. Daniel A. Farber*: Disaster Law,* Edward Elgar, 2010.
4. Daniel A. Farber*: Disaster Law and Policy,* Aspen Publishers, 2010.
5. Arun Ray, Mohapatra: *Public Interest Litigation and Human Rights in India, Delhi,* Radha Publications, 2001.

ADCDM -VI: Project Report Based on Field Work.

Maximum Marks : 100

- Paper ADCDM-II will be based on Project Report
- Each student will be required to prepare a project work of 50-70 typed pages and submit it at least one week before the commencement of the annual Practical Examinations.
- The Project Report will be evaluated by an internal as well as external examiner.

*****